

Multimodal PET/MR Diagnosis with Variational Inference

— Experiment Report

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Abstract

We present an experimental evaluation of a multimodal variational autoencoder (MoPoE-VAE) for Alzheimer’s Disease diagnosis using PET/MR imaging. Three experiments compare different latent dimensions and loss configurations. The 256-dim latent space achieves the lowest training loss (0.670), outperforming the baseline (0.851) and the no-cross-modal variant (0.764).

1 Dataset

ADNI dataset: **225 subjects** (223 with T1 MRI, 119 FDG-PET, 62 Tau-PET). **38 subjects** have all three modalities. Total: 10.6 GB.

2 Method

MoPoE-VAE with modality-specific 3D encoders, product-of-experts fusion, and diagnosis classifier. Trained with reconstruction, cross-modal, KL divergence, classification, and contrastive losses.

3 Experiments

Table 1: Experiment Configurations

Config	Latent Dim	Cross-Modal	Contrastive
Exp1 (Baseline)	128	Yes	Yes
Exp2 (Large)	256	Yes	Yes
Exp3 (No-Cross)	128	No	No

Table 2: Training Results (20 epochs, synthetic data, RTX 4060)

Experiment	Epoch 0	Epoch 20	Best	Time
Exp1 (Baseline)	2.495	0.864	0.851	220s
Exp2 (Large)	2.478	0.685	0.670	230s
Exp3 (No-Cross)	2.526	0.764	0.763	215s

3.1 Analysis

Latent Dimension: Increasing from 128 to 256 improves loss by 21% (0.851→0.670). The larger latent space provides more capacity for multimodal fusion.

Cross-Modal Loss: Removing cross-modal and contrastive losses reduces total loss (0.764 vs 0.851). However, this comparison is confounded by different loss compositions. On real data, cross-modal synthesis should improve diagnosis when modalities are missing.

Pretrained Weights: No directly applicable pretrained weights exist for our MoPoE-VAE architecture. MONAI segmentation models could be adapted for encoder initialization through transfer learning.

4 Conclusion

The 256-dim MoPoE-VAE (Exp2) achieves the best training performance. All three experiments validate the training pipeline. Full ADNI dataset ready for real-data experiments.

5 Repository

<https://github.com/tzhazuma/bmebrainproject>